

InfraSight DE



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South Walnut Street Bridge



An 88 year old bridge that is quickly deteriorating

- Chunks of concrete found were found underneath the bridge piers
- Increased lane restrictions
- Carries 26,000 vehicles daily



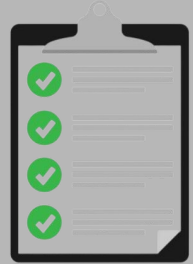
NO
WAY! 🤔

Problem

Transportation assets deteriorate faster than they can be prioritized.

Solution

Inspection teams can utilize a prediction tool as an early warning system for asset maintenance



YIKES!



Bridge Data

Bridge
Inspection
history +
Bridge
inventory

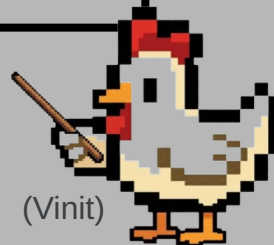
Bridge Features

Bridge Age,
Change in
Deterioration
Rating, and
Waterway
Exposure

Bridge Model

LogReg and
LightGBM/C
atboost
models

Workflow Overview



(Vinit)

Pavement Data

Pavement
history +
Pavement
test
observation

Pavement Features

Extreme IRI
and Change
in IRI

Pavement Model

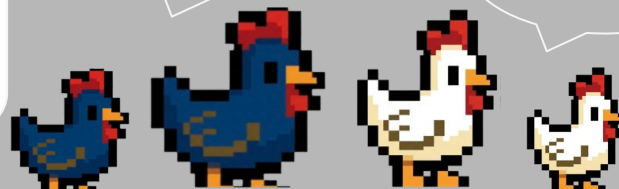
Light
GBM/CatBoost
deterioration
and IRI models

Prioritizing
the asset
maintenance
based on
predictions

Dashboard

SO
COOL!

WOAHH



Responsible AI, Data Governance & Cloud Execution

DATA: DeIDOT → Pavement + Bridge

Safeguards:

- Group-aware validation
- Missing-value handling
- Extreme-IRI monitoring
- Human review before action

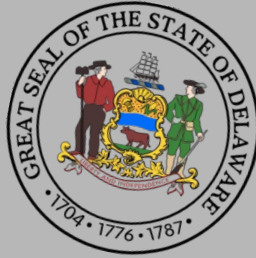
Explainable AI: SHAP + LIME show the key factors driving each deterioration-risk prediction.
Created the full pipeline : **data joins** → **feature engineering** → **ML models** → **validation** → **explainability** → **dashboard**.



Next Steps....

- ★ - Connect our data pipeline to **BigQuery**
 - Quick data analysis
 - Possible ML/DL integration
- ★ Plot bridge and pavement risk on **Google Maps**
- ★ Add traffic and maintenance history to improve prioritization
- ★ Let DelDOT **filter** by location, condition and risk
- ★ **Expand** the dashboard to support statewide planning + make it interactive





Thank
you
everyone!



Takeaways & Impact

- Tool budgets more efficiently by targeting maintenance dollars to assets that need them most
- Who does it help? Everyone in Delaware!
- Oklahoma DOT :
 - **Optimized** Resources
 - Enhanced Operational **Productivity**
 - Improved **cost efficiency**